SPECIFIC AIMS

Breathing behavior in resting conditions consists of active inspiration (inhalation) and passive expiration (exhalation). During periods of higher respiratory demand, expiration goes from passive to active to increase ventilation. The rhythm for active expiration emanates from a multifunctional region dubbed the parafacial respiratory group (pFRG), wherein its lateral margin, i.e., the lateral parafacial region (pF_L) is rhythmogenic[1]. The cellular composition of the pF_L is unknown and its borders are fuzzy. Nevertheless, the pF_L is thought to be anatomically and functionally distinct from the adjacent and partially overlapping chemosensitive neurons of the ventral parafacial region (pF_V), also known as the retrotrapezoid nucleus (RTN)[2,3]. The molecular genetics, ontology, physiology, and function of pF_V neurons are well defined[4–6] but not so for the pF_L. Our overall objective is to characterize the neuron population that constitutes the expiratory breathing oscillator in the pF_L.

OUR STRATEGY:

- Sequence the transcriptomes of parafacial interneurons;
- Identify transcripts that define neuronal subtypes and map their spatial location; and
- Interrogate expiratory function by activating and/or silencing molecularly defined pF_L neuron populations in awake behaving mice.

AIM 1: SEQUENCE THE TRANSCRIPTOMES OF THE PFRG INTERNEURONS. We will test the hypothesis that pF_L expiratory rhythmogenic neurons have a distinct molecular fingerprint that uniquely differentiates them from other parafacial cells, including chemosensitive neurons[7] and glia[8,9], as well as the C1 cells[10]. The presumptive pF_L can be loosely defined by peptide receptor expression, glutamatergic transmitter phenotype, and the lack of *Phox2B* expression[11–14]. We will utilize the droplet-based 10x Chromium Single Cell Gene Expression with feature barcoding technology to create a single-cell cDNA library from cell suspensions obtained from tissue sections of the pFRG. We will sequence their transcriptomes and apply genomic data science to define expression profiles for pFRG cell types and identify clusters with similar profiles.

AIM 2: DEFINE THE PUTATIVE CELLULAR COMPOSITION OF THE PF_L. We will test the hypothesis that specific transcripts, or combinations of transcripts, which putatively define cell clusters (Aim 1) can be used to map cell populations in the pF_L. We will identify these specific transcripts histologically using RNAScope HiPlex v2 assay, an *in situ* hybridization (ISH) tool that detects gene expression patterns in fixed-tissue sections at single cell resolution. We will visualize all the major cell types and/or clusters identified in Aim 1 and determine their relative spatial distribution to establish the borders of the putative pF_L and its constituent neurons.

AIM 3: INTERROGATE THE EXPIRATORY RHYTHMOGENIC ROLE OF PF_L INTERNEURONS. We will test the hypothesis that a molecularly defined neuron type underlies active expiration. We will use a virus-based approach to install optogenetic proteins in specific neuron classes identified in Aim 2. We will photostimulate or photoinhibit putative pF_L interneurons and observe attendant effects on active expiration, which we anticipate will establish their functionality.

DELIVERABLE: We will define the cellular core of the expiratory breathing oscillator in terms of molecular genetics. We will also define its anatomical borders. Doing so provides new information about its neural development and enables us to perturb its function and unravel its mechanisms of operation.

IMPORTANCE: The breathing central pattern generator (bCPG) consists of at least two, perhaps three, oscillatory microcircuits that generate the rhythm for breathing movements[1,15–19]. The inexorable inspiratory oscillator is well understood at the cellular and molecular level and its borders are well defined. By contrast the expiratory oscillator is opaque and its cellular composition remains fuzzy. This project would provide new knowledge to define the constituent neurons of expiratory pF_L oscillator and their functionality. The applications of this knowledge pertain to understanding the neural origins of breathing behavior and bCPG development.

SIGNIFICANCE

FILLING A KEY KNOWLEDGE GAP IN MOTOR CONTROL: Inspiration is the inexorable active phase of breathing, and its neural origins are relatively well understood at the level of cells and synapses, their physiological properties, as well as their molecular-genetic origins. Expiration, by contrast, is a mess of conflicts. Consider that expiration itself is biphasic, being subdivided into post-inspiration and late expiration[1,15,16,18]. Some posit that post-inspiration comes from a conditional brainstem oscillator[19] whereas others consider post-inspiration an elaboration of the motor pattern for inspiration that involves largely pontine circuits[20,21]. Regarding late expiration, we know that it becomes active through disinhibition of GABA- and glycinergic interneurons[22–24]. Disinhibition comes about via a microcircuit in the Bötzinger complex[22]. Note: the Bötzinger region is similar only in name – because it is adjacent to the preBötC and was discovered first[25–27] – but it is neither inspiratory nor rhythmogenic in any sense. Regarding the mechanism underlying expiratory rhythm all we know is that rhythm-generating neurons are located at the lateral edges of the facial cranial nucleus (VII), i.e., the region dubbed pF_L[3,12,23,28]. The borders of this putative expiratory region pF_L, the molecular genetic origins of pF_L neurons, and their physiological properties remain unknown. Few – if any – researchers are currently addressing the unknown borders of the pFL or its cellular constituents. Here we will characterize the cellular origins of expiratory breathing-related behavior in mammals.

SIGNIFICANCE FOR SENSORIMOTOR NEUROSCIENCE: Rhythmic movements in all animals emanate from a type of central pattern generator (CPG) network[29]. Three CPGs in mammals are well described: the whisking CPG in rodents, the oral-motor CPG, and the inspiratory core of the bCPG. What is striking about these three CPGs is their widely divergent mechanisms: the whisking CPG depends on relatively fast (>10 Hz) oscillations that emerge via mutually inhibitory synaptic interactions[30,31]. The oral-motor CPG depends on astrocyte-mediated extracellular Ca^{2+} chelation that induces rhythmogenic pacemaker neurons[32]. The inspiratory core CPG depends on emergent network properties among excitatory (glutamatergic) interneurons without need of either synaptic inhibition or neuronal pacemaker properties[1,17,33,34]. Elucidating the cellular core of the expiratory oscillator will likely reveal a unique rhythmogenic mechanism that advances sensorimotor neuroscience.

SIGNIFICANCE FOR UNDERSTANDING BCPG DEVELOPMENT: During development the neurons that form the chemosensors of the pF_V express transcription factors *Atoh1* embryonically and *Phox2b* postnatally[35–38]. The embryonic and early postnatal *Atoh1/Phox2b* neurons exhibit CO_2-sensitivity as well as phasic burst activity during post-inspiration as well as late expiration[14,37,39,40]. But by adulthood their expiratory phasic activity is gone and the *Phox2b*-expressing neurons only fire tonically with a frequency that encodes CO_2 (thus pH) levels[4,7]. What happens to expiratory bursting? There are two possibilities: First, *Atoh1* controls a superset of interneurons that diverge to form *Phox2b*$^+$ chemosensors and *Phox2b*$^-$ expiratory rhythmogenic neurons. Early in development the superset expresses both chemosensory and expiratory properties. Second, expiratory rhythmogenic neurons do not derive from *Atoh1*-expressing precursors but project to them early in development and thus impart expiratory rhythmicity (presumably those synaptic connections get pruned during maturation). Both hypotheses are viable because the chemosensors and expiratory rhythm generators emerge from the rhombomeres 3 and 5 (r3/r5) microcircuits distinguished by *Krox20* (i.e., *Egr2*) expression during early embryonic development[41].The intermingling of expiratory and chemosensory functions in *Atoh1/Phox2b* neurons has confounded the field for decades. The present project could elucidate this developmental mystery and thus advance our understanding of bCPG development.

TRANSLATIONAL SIGNIFICANCE OF THE PROPOSAL: Brain-related breathing disorders like congenital central hypoventilation syndrome (CCHS) and Pitt-Hopkins syndrome act on the pF_V/RTN and cause chemosensory neuron dysfunction[6,35,42,43]. Some forms of sudden infant death syndrome (SIDS) may in part be attributable to dysfunction of serotonergic neurons of the brainstem raphe, which lowers excitability throughout the bCPG leading to hypopnea and apnea[44,45]. Maladies like apnea of prematurity[46], opioid-induced respiratory depression[47,48], and respiratory failure in neurodegenerative diseases act, in part, on the preBötC and affect the ability to generate and sustain inspiratory rhythm[49–52]. This project would address the expiratory component of the bCPG whose constituent precursor cells express *Krox20* (i.e., *Egr2*) during embryonic development and form an opioid-insensitive rhythmogenic circuit at or around the time of birth[41,53]. Understanding the cellular origins of opioid-insensitive expiratory rhythms may be applicable to treatment and prevention strategies that protect recreational and clinical opioid drug users, i.e., both licit and illicit uses of opioid drugs can depress breathing.

KNOWLEDGE ADDED TO THE COMMONS. We will sequence the transcriptomes of pFRG neurons (Aim 1) and release all raw and processed data via National Center for Biotechnology Information (NCBI) Gene Expression Omnibus (GEO) because we value openness, sharing, and reproducibility. These genomic data can then be used by other respiratory physiologists and neuroscientists for any form research including translation.

INNOVATION

INTELLECTUAL INNOVATION: Usually neuroscientists locate the cellular core of a CPG via serial transection experiments, followed by years of cellular electrophysiology using reduced slice preparations in conjunction with neuroanatomy (immunohistochemistry and *in situ* hybridization). Here we turn that discovery model upside down: we start with genetic information and use data science to develop candidate populations followed by anatomical mapping and optogenetic interrogation of said populations in living breathing mice. This approach alleviates the need for decades of cellular electrophysiology and gets us the answers we seek rapidly. Please note we are **not** disparaging electrophysiology (PI Del Negro is classically trained in the discipline.) Rather, by exploiting genomic data science and *in vivo* optogenetics we can obtain definitive new knowledge quickly.

TECHNICAL TRIFECTA: First, we exploit single-cell barcoding, next-generation sequencing, and genomic data science to analyze the transcriptomes of thousands of pFRG cells, some of which are the elusive expiratory rhythmogenic core of the pF_L. Second, we harness multi-plex in situ hybridization (RNAScope HiPlex) to anatomically locate the population and define pF_L borders. Third, we use virus delivery of optogenetic tools to test behavioral function of the pF_L constituent neurons. Although optogenetics is now a common tool in neuroscience, it is nonetheless powerful and appropriate. The genomic data science and multi-plex *in situ* hybridization are still appropriately considered “cutting edge” and innovative. The marriage of data science and anatomical techniques with optogenetic studies *in vivo* is technical trifecta.

INVESTIGATORS

The applicants are faculty from Applied Science, Neuroscience, and Biology whose overlapping expertise span molecular biology, *in vivo* experimental neuroscience, applied mathematics, and bioinformatics. Our collaborative team has produced three publications since 2017 using RNA-Seq, Patch-Seq, and genomic data science[54–56]. We disseminated our data in the public domain: NCBI GEO Accession Nos. GSE100356, GSE175642, & GSE175643. PI Del Negro has 23+ years of experience in respiratory neurobiology.

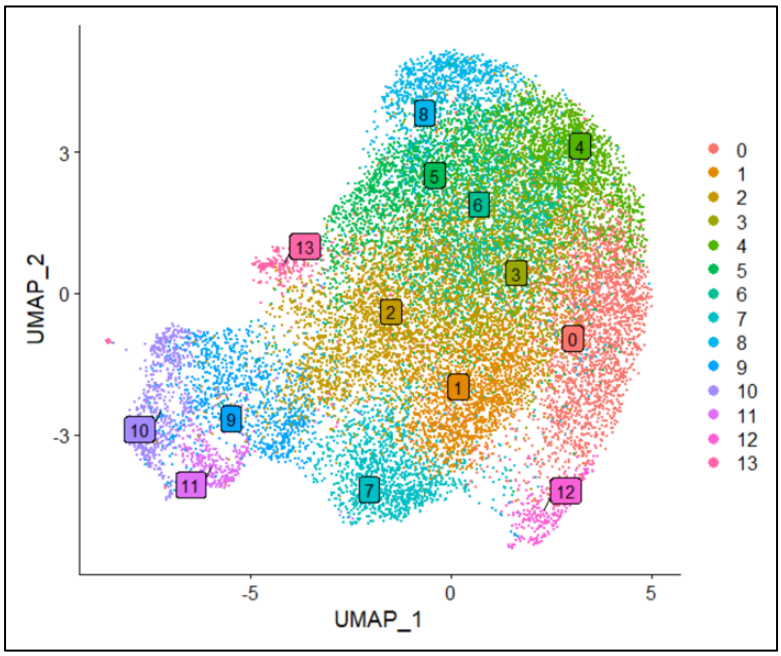
APPROACH

AIM 1: SEQUENCE THE TRANSCRIPTOMES OF THE PFRG INTERNEURONS. We will perform single cell RNA-Sequencing (scRNA-Seq) and single nucleus RNA-Sequencing (snRNA-Seq) on dissociated cells from adult mouse pFRG to identify and characterize the neuron population that putatively constitutes the expiratory breathing oscillator pF_L.

Because relatively little is known regarding the molecular composition and identity of the expiratory breathing oscillator neurons, a single cell sequencing strategy is the only approach that can identify cell types, particularly novel ones, in a relatively unbiased manner. scRNA-Seq allows for the identification of meaningful clusters of cell types, rare or unique cell types, cellular heterogeneity and relatedness, and potential co-regulated gene regulatory networks that define the various cell types. The rationale for employing both scRNA-Seq and snRNA-Seq is that given the foundational nature of this first aim – to produce the most accurate and comprehensive dataset of the pFRG to then discover the constituent expiratory rhythmogenic pF_L neurons – scRNA-Seq and snRNA-Seq are expected to produce distinct and complementary outcomes. scRNA-Seq typically returns larger numbers of transcripts and reflects the acute cytoplasmic state of the cell as it pertains to RNA stability. In contrast, snRNA-Seq avoids transcriptional changes like RNA degradation that occur as a result of tissue dissociation. Therefore, snRNA-Seq returns a more uniform inclusive RNA sample of cells from tissues that are particularly challenging to dissociate.

Adult mice will be deeply anesthetized and transcardially perfused with cold phosphate-buffered saline solution (PBS). Next, we will perform cervical dislocation, cut the head, and remove the brain and brainstem. The pF_L will be dissected in a dissociation enzyme and transcription inhibitor included in the Adult Brain Dissociation Kit (Miltenyi Biotec, Gaithersburg, MD). Brains will be kept ice-cold during dissection of all samples (total duration <2 hr). For the scRNA-Seq, we will follow 10X protocols that has been widely and successfully used in the published literature, and in our pilot experiments. Briefly, following dissociation, cellular debris will be removed via centrifugation and filtering, counted, and resuspended at 700-1200 cells/μl before proceeding immediately

to the 10X Single Cell protocol. For nuclei isolation we will follow the well-vetted 10X Genomics protocol for nuclei isolation from adult mouse brain tissue. Briefly, following successful lysis of cells with fresh lysis buffer, myelin will be removed using commercial myelin removal beads, and the nuclei will be washed, resuspended, counted, and then purified further using sucrose density gradient centrifugation. Nuclei will be resuspended using the recommended 700-1200 nuclei/µl concentration. The nuclei and cells will be directly used in the 10X Single Cell protocol, which requires 10,000 cells per run of scRNA-Seq and 10,000 nuclei per run snRNA-Seq. We will perform 3 runs of each approach, so 6 runs total. To acquire adequate starting material for a run, we need N = 17 mice, as we explain below. The number of neurons that comprise the pFRG is unknown, but we estimate its size to be ~2000 cells, neurons and glia included. So, if droplet capture worked perfectly, then we could acquire enough starting material from N = 10 mice. However, there is a 70% loss rate associated with droplet capture – i.e., 30% recovery rate – so we only get 600 cells per mouse. So, to acquire the 10,000 cells or 10,000 nuclei for a run we need N = 10,000 ÷ 600 ≈ 17 mice per run. Considering 3 runs of scRNA-Seq and 3 runs of snRNA-Seq approach and N = 17 mice per run, the Aim requires a total of N = 17 X 6 = 102 mice. Libraries will be sequenced on an Illumina NovaSeq6000.



Next-generation sequencing returns FASTQ files, which we will quality check using FastQC among other tools. For each sequencing run, genes-of-origin will be determined by aligning reads to the mm10 reference murine genome (UC Santa Cruz genome browser). Reads will be sorted by the cell barcodes, and unique molecular identifiers will be counted for each gene in each cell. The resulting expression matrix (approximately 30,000 genes across ~1,000 cells) will have log(x+1) transformed entries where x is number of transcripts[57].

We will characterize cell diversity through *ab initio* reconstruction of cell types. Controlling for the relationship between mean expression and variability, we will identify the genes that are most variable. Using unsupervised dimension reduction via principal component analysis (PCA) on highly variable genes and randomization using the R package Jackstraw, we will identify statistically significant PCs for further analysis[58]. These results will be used as input for t-distributed stochastic neighbor embedding (t-SNE)[59], uniform manifold approximation and projection (UMAP)[60], and other cluster analysis methods to obtain a two-dimensional embedding of pFRG single-cell expression profiles. Distinct cell types should form point clouds within the t-SNE and UMAP projections (Fig. 1), one of which will most likely represent the pF_L. Putative cell types will be identified using density clustering, e.g., as implemented in the DBSCAN package in R[61]. Post-hoc tests searching for differentially expressed genes will ensure that identified clusters are not over-partitioned and, thus, represent distinct cellular classifications[62]. Results obtained using custom R/python scripts will be compared to Cell Ranger Analysis pipeline.

We have successfully performed scRNA-Seq using presumptive neural tissue from *Xenopus laevis* embryos, completing the entire pipeline from tissue dissection through data analysis using the 10X platform (Fig. 1). These skills will be immediately and directly applicable to our project examining the pFRG to find the pF_L.

Anticipated outcomes include providing the first molecular characterization of the putative expiratory oscillator and revealing transcriptional similarities and differences within the population of cells in the pF_L. Studying co-expression patterns at the single cell level will allow identification of genes and gene networks that underlie cell-type specification and functional heterogeneity. To identify candidate genes to be utilized in Aims 2 and 3, we will prioritize transcripts whose expression are in agreement in both scRNA-seq and snRNA-seq. At minimum, we aim to identify clusters of glutamatergic and possibly a small cluster of GABA/glycinergic neurons. We also anticipate that a specific subset of peptides or peptide receptors could define the pF_L; peptides and receptors have been invaluable in unraveling the constituent cells and borders of the preBötC. We hope to uncover a transcription factor, like *Phox2B*, that can be used to manipulate cellular activity (Aim 3).

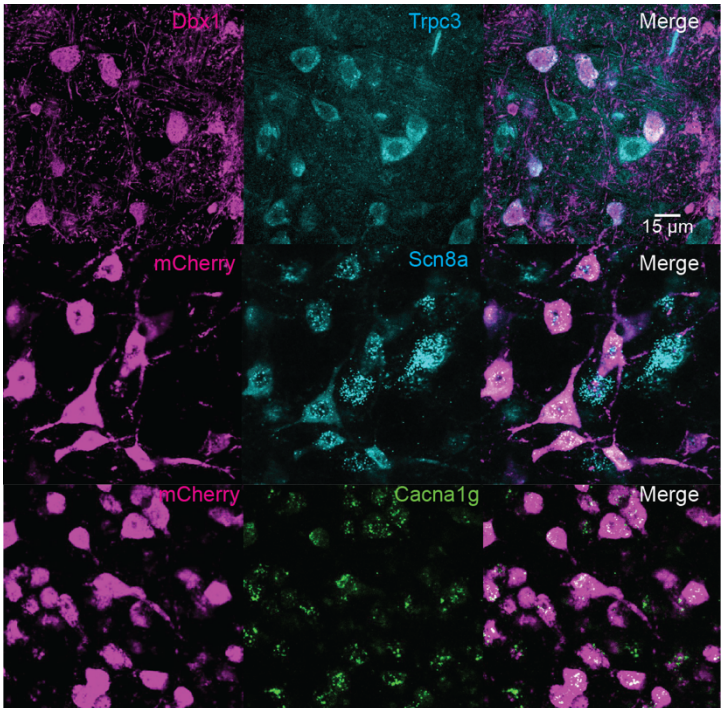


***Figure 2**. histology in preBötzinger complex respiratory neurons. Top: Dbx1, Trpc3, and merged images from immunocytochemistry. Middle: Vglut2 (mCherry), Scn8a ($Na_V1.6$), and merged images. Lower: Vglut2 (mCherry), Cacna1g ($Ca_V3.1$), and merged images.*

AIM 2: DEFINE THE PUTATIVE CELLULAR COMPOSITION OF THE pF_L. Analyzing transcriptomic profiles of single cells by scRNA-Seq and snRNA-Seq (i.e., Aim 1) is now a widespread tool that can identify novel cell populations and help understand their role(s) and function(s). This has proven to be especially true in complex organs whose constituent cells are heterogeneous, dispersed, and numerous – such as the mammalian brain. The downside to scRNA-Seq is that dissociated cells constitute its input, which divorces cells from their respective anatomical sites; information about spatial organization is lost. As a remedy, confirmation and spatial mapping of scRNA-Seq results can be obtained using assays that retain spatial organization, such as RNA *in situ* hybridization.

We will use multiplex *in situ* hybridization (commonly known as RNA-Scope, Advanced Cell Diagnostics, Newark, CA) to quantify transcripts in the pF_L region and validate genetic signatures with spatial context at the single cell level.

We have a great deal of experience with histology to characterize protein and transcript expression in respiratory neurons from adult and neonatal mice. For example, we examined expression of nonselective cation channels Trpm4 and Trpc3 at the protein level in preBötC rhythmogenic Dbx1-derived neurons using immunolabeling (Fig. 2, top row)[63]. We routinely utilize RNA-Scope to quantify ion channel transcripts in the context of short hairpin RNA (shRNA) gene-knockdown experiments. The middle and lower rows of Figure 2 show images from RNA-Scope multiplex experiments on adult mice using probes that target *Vglut2* (*Slc17a6*) shown in mCherry, which identifies the same Dbx1-derived glutamatergic preBötC population, as well as *Scn8a* ($Na_V1.6$) and *Cacna1g* ($Ca_V3.1$) transcripts. The RNA-Scope procedure not only provides positional information of the neurons expressing these transcripts but also allows visualization of individual transcript molecules (middle panels) for quantification. Although we do not yet have pilot data for putative pF_L neurons (it would be foolish to try and acquire it before having completed Aim 1) we are confident that our extensive experience with RNA-Scope in the rhythmogenic interneurons of the preBötC will be immediately applicable to studying the pF_L neurons after Aim 1 helps us identify the appropriate target transcripts.

We will evaluate the expression of the mRNA in the pF_L of adult animals, because doing so matches the age and developmental stage as the tissue in Aim 1. Adult mice will be deeply anesthetized with isoflurane, then transcardially perfused with PBS, followed by 4% paraformaldehyde. 30 µm-thick brain sections will be cut using a vibratome and then mounted onto super-frost slides. Brain sections will be pretreated in a series of steps involving fixation, drying, alcohol dehydration and target recovery. Next, RNA-Scope reagents will be used to label the mRNA molecules of the target transcripts. We custom design or purchase commercially available probes to target mRNA molecules in our *in situ* RNA-Scope experiments. We are adept at probe design so this would pose no problem if our Aim 1 results return an exotic gene for which there is no commercially available probe.

Fluorescent Opal dyes and DAPI solution will be used for visualizing the transcripts and cell nuclei. ProLong Gold Antifade mounting agent (Invitrogen, Waltham MA) will be used to preserve the fluorescent signal with a glass coverslip. Sections will then be visualized under the four-channel, upright confocal microscope body based on Cerna System (Thorlabs, Newton NJ), and images will be acquired using the ThorimageLS data acquisition software. Quantification of mRNA transcripts will be assessed using the open-source software Quantitative Pathology (QuPath)[64].

We expect to find marker genes for pF_L neurons that differentiate them from nearby pF_V neurons, C1 neurons and Facial (VII) motoneurons: pF_V neurons express *Phox2b* and they appear to concentrate at the ventral area

of the pFRG, just outside of the pyramidal tract. We expect to clearly demarcate the ventral border of the VII cranial nucleus via motoneuron-specific genes like *Chat* (ChAT). C1 neurons also express *Phox2b* as well as tyrosine hydroxylase (Th)[10]. Having at least three distinct markers, we expect, will well-define the borders of the expiratory oscillator pF_L.

As an alternative approach we may focus on peptides and peptide receptors, which have proven to be critical in defining the borders of the preBötC inspiratory oscillator[65–69]. We already know that neurokinin-1 receptors (NK1Rs) are expressed in the putative pF_L[12,28] but like in preBötC that marker is promiscuous and is very unlikely to differentiate the pF_L from the pF_V, whose constituent neurons also express NK1Rs[70,71]. Expiratory rhythm in neonatal mice is opioid-insensitive[53,72], and may also be so in adults[73], so the μ-opioid receptor is unlikely to be a good marker for the pF_L. Bombesin-like peptides GRP and NMB also appear to be exclusively associated with the pF_V and thus unlikely to differentiate the pF_L from pF_V[74,75]. The list above is nonetheless useful because the expression of those peptides can be used to exclude cells from the pF_L. We speculate that peptides not on the list above, but identified in cell clusters in Aim 1, might be key markers for the pF_L.

AIM 3: INTERROGATE THE EXPIRATORY RHYTHMOGENIC ROLE OF pF_L INTERNEURONS. The cellular mechanisms that generate active expiration remain fuzzy. Expiratory breathing movements have been evoked by photostimulation in the pFRG, which certainly contains the pF_L,[23] but a valid critique is that those investigators used the generic neuronal promoter synapsin (SYN) to drive ChR2 expression and their illumination strategy covered an enormous surface area over the pFRG and regions adjacent to it. Therefore, the study provides little insight into the borders or cellular composition of the expiratory oscillator. Another study evoked active expiration indirectly by photostimulation of the pF_V[76], which simulates hypercapnia (elevated respiratory demand) and elicits active expiration. So, again, that study provides no insight into the pF_L. Other groups evoked active expiration via focal disinhibition, accomplished by locally microinjecting $GABA_A$ and glycine receptor antagonists[12,13,22,23,77]. Again, the spillover of drugs into adjacent regions cannot be avoided, which undeniably introduces ambiguity regarding which population(s) were affected. Therefore, neither the prior optogenetic studies nor studies using disinhibition helped in any way to determine the characteristic population underlying active expiration or to establish the borders of the pF_L.

Here we would advance knowledge of expiratory function at the cellular level by exploiting the discoveries from Aims 1 and 2 to express opsins, ChR2 and the hyperpolarizing proton pump Archaerhodopsin (Arch), in pF_L neurons to interrogate their role. To definitively establish expiratory rhythmogenic function, one must show that selective photostimulation of pF_L neurons directly induces active expiration as well as show that during endogenous conditions for active expiration (e.g., hypercapnia) that expiratory function is abrogated by photoinhibition that silences the pF_L neurons.

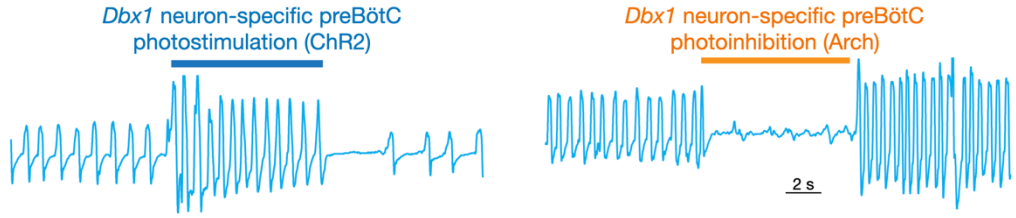


***Figure 3**. whole-body plethysmography showing photostimulation and photoinhibition of preBötC inspiratory rhythmogenic neurons in adult mice.*

We will exploit those transcript markers that emerge in Aims 1 and 2. We would ideally adopt a strategy such as the one used to express opsins in chemosensory neurons of the pF_V, which involves the artificial *Phox2b*-specific promoter PRSx8. Likewise, if we discover a transcription factor active in adult pF_L neurons then we would similarly design an artificial promoter to express the opsin in a cell-specific manner after viral transfection. This approach would be equally applicable for photoinhibition, the only change would be to include Arch instead of ChR2 in the expression vector.

In the unlikely event that we do not discover a suitable transcription factor to exploit, our fallback plan would be to exploit transmitter phenotype, the most likely scenario being *Vglut2* expressed in putatively expiratory rhythmogenic neurons. We would use a standard Cre-dependent FLEX virus to install ChR2 and Arch in pF_L neurons of $Vglut2^{Cre}$ mice. We maintain a colony of these mice for other projects so our fallback plan would present no delay or setback. Even that fallback approach would be a dramatic improvement over the pharmacological disinhibition via injection or unselective photostimulation over gross surface area as in prior studies. So, to summarize, we would harness the marker(s) from Aims 1 and 2 to selectively target pF_L

neurons, but even if that does not work, we would still be able to manipulate putative expiratory rhythmogenic neurons by selectively targeting them based on transmitter phenotype within Aim 2-defined pF_L borders.

We will deeply anesthetize adult mice, expose the skull, perform 0.5 mm diameter craniotomies 5.8-6.2 mm posterior to bregma and 1.5-1.8 mm lateral to the midline suture at a depth of 5.5-5.9 mm. We will inject adeno-associated virus (AAV) carrying expression vectors for either ChR2 or Arch (or a more powerful opsin, should it emerge). The nature of the promoter, and whether viral expression is Cre-dependent will depend on the outcome of Aims 1 and 2. Our experience with genetic and reverse genetic technologies assures us that we will be able to harness the marker genes or transmitter phenotype to install the opsins appropriately. A micro-syringe carrying the viral vector will descend at 10 μm/s. Once in place, we will inject at ~60 nl/min for at least 2 min. The syringe will remain in place for 10 min and be retracted at 10 μm/s. We are not particularly worried about spillover because we expect to have a specific marker for the pF_L that would diminish the likelihood of expression in adjacent VII motoneurons, pF_V chemosensors, or C1 neurons.

During the same surgery, after viral injection, the mice will be equipped with fiber optic appliances constructed by joining 1.27-mm diameter ceramic ferrules (Precision Fiber Products, Chula Vista, CA) with 105-μm diameter 0.22 numerical aperture (NA) multimode fibers (Thorlabs). The fiber will be secured with a cyanoacrylate adhesive (Loctite 3092, Henkel Corp., Rocky Hill, CT) We will inject virus and implant optical fibers unilaterally for photostimulation, but we will do so bilaterally for photoinhibition. Unilateral ChR2 activation may be sufficient to evoke active expiration while making the surgery simpler and easier from which to recover. Bilateral Arch activation may be necessary to arrest the behavior, thus justifying the need for more complicated surgery and recovery. We worked out this strategy for optogenetics in the preBötC[78,79] (Fig. 3).

Breathing will be measured using whole-body plethysmography, specifically using a pneumotachograph to measure airflow, which facilitates detection of active expiration. While barometric whole-body plethysmography is generally preferred for breathing measurements because of its accuracy regarding tidal volume, the barometric technique would also require abdominal electromyography (EMG) to detect active expiration. Inclusion of abdominal EMG would necessitate deep anesthesia, which augments inhibitory synaptic drive to the pFRG and suppresses active expiration. We prefer to do the experiments in awake adult mice with a pneumotachograph and accept the limits regarding tidal volume.

After acclimating to the plethysmograph mice typically exhibit epochs of eupnea (quiet unlabored breathing) lasting 20-30 s. During such bouts we will photostimulate using 473 nm light at 6-10 mW for 2 s. Those strengths and durations were effective when photostimulating the preBötC of awake mice, but are shorter than 10-s pulses used earlier on anesthetized rats in the pFRG[23]. We prefer the shorter 2-s duration to minimize conscious perturbation of the mouse's behavior and state of mind. If we do not observe a phenotype, then we can always increase pulse duration further. We will identify active expiration by outward airflow immediately prior to inspiration, i.e., during late expiration, as shown in Fig. 4. During eupnea there is no active expiration; but we predict that illumination will evoke it.

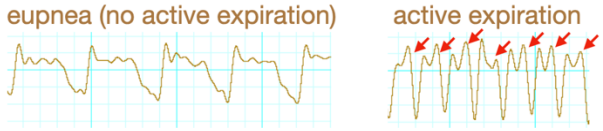


***Figure 4**. Peak in airflow immediately prior to inspiration (red arrows, right) are the hallmark feature of active expiration.*

In the next set of experiments we will expose mice to mildly hypercapnic conditions, e.g., end-tidal CO_2 ~6-8%, enough to observe active expiration but not enough to induce dyspnea[80,81]. Once the mice are acclimated and exhibit active expiration during most of their breaths, we will activate Arch using 6-10 mW pulses of 589 nm light for a duration of 2-10 s. Here we will adjust the duration of the pulse to maximize the phenotype; 2 s may be sufficient. We predict that photoinhibition should diminish or prevent active expiration in hypercapnia.

SUMMARY. Breathing may the first rhythmic motor behavior whose neural origins are understood at multiple levels of analysis in the nervous system: from genes to ion channels, to cells and networks. Presently, there is a large lacuna in that the constituent neurons and borders of the expiratory oscillator are unknown whereas the inspiratory oscillator, the central chemosensors, the chemo- and mechano-sensory afferents, as well as pontine sites are understood at a very high level. This project would rectify that knowledge imbalance and provide key new insights that have developmental and translational significance for medicine and neuroscience in general. The team is an interdisciplinary one with expertise that span molecular biology, experimental behavioral neuroscience, genomic data science and applied mathematics. The team already has track record of productivity with genomic data sets released to the public domain.

BIBLIOGRAPY AND REFERENCES CITED

1. Del Negro, C. A., Funk, G. D. & Feldman, J. L. Breathing matters. *Nat. Rev. Neurosci.* **19**, 351–367 (2018).
2. Moreira, T. S. *et al.* The retrotrapezoid nucleus and the neuromodulation of breathing. *J. Neurophysiol.* **125**, 699–719 (2021).
3. Pisanski, A. & Pagliardini, S. The parafacial respiratory group and the control of active expiration. *Respir. Physiol. Neurobiol.* **265**, 153–160 (2019).
4. Guyenet, P. G. *et al.* The Retrotrapezoid Nucleus: Central Chemoreceptor and Regulator of Breathing Automaticity. *Trends Neurosci.* **42**, 807–824 (2019).
5. Guyenet, P. G. & Bayliss, D. A. Neural Control of Breathing and CO2 Homeostasis. *Neuron* **87**, 946–961 (2015).
6. Amiel, J. *et al.* Polyalanine expansion and frameshift mutations of the paired-like homeobox gene PHOX2B in congenital central hypoventilation syndrome. *Nat. Genet.* **33**, 459–461 (2003).
7. Mulkey, D. K. *et al.* Respiratory control by ventral surface chemoreceptor neurons in rats. *Nat. Neurosci.* **7**, 1360–1369 (2004).
8. Angelova, P. R. *et al.* Functional Oxygen Sensitivity of Astrocytes. *J. Neurosci. Off. J. Soc. Neurosci.* **35**, 10460–10473 (2015).
9. Gourine, A. V. *et al.* Astrocytes control breathing through pH-dependent release of ATP. *Science* **329**, 571–575 (2010).
10. Guyenet, P. G. *et al.* C1 neurons: the body's EMTs. *Am. J. Physiol. Regul. Integr. Comp. Physiol.* **305**, R187-204 (2013).
11. Biancardi, V. *et al.* Mapping of the excitatory, inhibitory, and modulatory afferent projections to the anatomically defined active expiratory oscillator in adult male rats. *J. Comp. Neurol.* cne.24984 (2020) doi:10.1002/cne.24984.
12. Huckstepp, R. T. R., Cardoza, K. P., Henderson, L. E. & Feldman, J. L. Role of Parafacial Nuclei in Control of Breathing in Adult Rats. *J. Neurosci.* **35**, 1052–1067 (2015).
13. de Britto, A. A. & Moraes, D. J. A. Non-chemosensitive parafacial neurons simultaneously regulate active expiration and airway patency under hypercapnia in rats: Hypercapnia and active expiration. *J. Physiol.* **595**, 2043–2064 (2017).
14. Onimaru, H., Ikeda, K. & Kawakami, K. CO2-Sensitive Preinspiratory Neurons of the Parafacial Respiratory Group Express Phox2b in the Neonatal Rat. *J. Neurosci.* **28**, 12845–12850 (2008).
15. Ramirez, J.-M. & Baertsch, N. Defining the Rhythmogenic Elements of Mammalian Breathing. *Physiol. Bethesda Md* **33**, 302–316 (2018).
16. Ramirez, J.-M. & Baertsch, N. A. The Dynamic Basis of Respiratory Rhythm Generation: One Breath at a Time. *Annu. Rev. Neurosci.* **41**, 475–499 (2018).
17. Ashhad, S., Kam, K., Del Negro, C. A. & Feldman, J. L. Breathing rhythm and pattern and their influence on emotion. *Annu. Rev. Neurosci.* **45**, 223–47 (2022).
18. Richter, D. W. & Smith, J. C. Respiratory rhythm generation in vivo. *Physiol. Bethesda Md* **29**, 58–71 (2014).
19. Anderson, T. M. *et al.* A novel excitatory network for the control of breathing. *Nature* **536**, 76–80 (2016).
20. Dhingra, R. R., Dick, T. E., Furuya, W. I., Galán, R. F. & Dutschmann, M. Volumetric mapping of the functional neuroanatomy of the respiratory network in the perfused brainstem preparation of rats. *J. Physiol.* **598**, 2061–2079 (2020).
21. Toor, R. U. A. S. *et al.* Neurons in the Intermediate Reticular Nucleus Coordinate Postinspiratory Activity, Swallowing, and Respiratory-Sympathetic Coupling in the Rat. *J. Neurosci. Off. J. Soc. Neurosci.* **39**, 9757–9766 (2019).

22. Flor, K. C., Barnett, W. H., Karlen-Amarante, M., Molkov, Y. I. & Zoccal, D. B. Inhibitory control of active expiration by the Bötzinger complex in rats. *J. Physiol.* **598**, JP280243 (2020).

23. Pagliardini, S. *et al.* Active expiration induced by excitation of ventral medulla in adult anesthetized rats. *J. Neurosci. Off. J. Soc. Neurosci.* **31**, 2895–2905 (2011).

24. Barnett, W. H. *et al.* The Kölliker-Fuse nucleus orchestrates the timing of expiratory abdominal nerve bursting. *J. Neurophysiol.* **119**, 401–412 (2018).

25. Merrill, E. G., Lipski, J., Kubin, L. & Fedorko, L. Origin of the expiratory inhibition of nucleus tractus solitarius inspiratory neurones. *Brain Res.* **263**, 43–50 (1983).

26. Fedorko, L. & Merrill, E. G. Axonal projections from the rostral expiratory neurones of the Bötzinger complex to medulla and spinal cord in the cat. *J. Physiol.* **350**, 487–496 (1984).

27. Ezure, K. Synaptic connections between medullary respiratory neurons and considerations on the genesis of respiratory rhythm. *Prog. Neurobiol.* **35**, 429–450 (1990).

28. Huckstepp, R. T., Henderson, L. E., Cardoza, K. P. & Feldman, J. L. Interactions between respiratory oscillators in adult rats. *eLife* **5**, (2016).

29. Grillner, S. & El Manira, A. Current Principles of Motor Control, with Special Reference to Vertebrate Locomotion. *Physiol. Rev.* **100**, 271–320 (2020).

30. Deschênes, M. *et al.* Inhibition, not excitation, drives rhythmic whisking. *Neuron* **90**, 374–387 (2016).

31. Takatoh, J. *et al.* The whisking oscillator circuit. *Nature* **609**, 560–568 (2022).

32. Morquette, P. *et al.* An astrocyte-dependent mechanism for neuronal rhythmogenesis. *Nat. Neurosci.* **18**, 844–854 (2015).

33. Kam, K., Worrell, J. W., Janczewski, W. A., Cui, Y. & Feldman, J. L. Distinct inspiratory rhythm and pattern generating mechanisms in the preBötzinger complex. *J Neurosci* **33**, 9235–9245 (2013).

34. Kallurkar, P. S., Grover, C., Picardo, M. C. D. & Del Negro, C. A. Evaluating the Burstlet Theory of Inspiratory Rhythm and Pattern Generation. *eNeuro* **7**, ENEURO.0314-19.2019 (2020).

35. Rose, M. F. *et al.* Math1 is essential for the development of hindbrain neurons critical for perinatal breathing. *Neuron* **64**, 341–354 (2009).

36. Rose, M. F., Ahmad, K. A., Thaller, C. & Zoghbi, H. Y. Excitatory neurons of the proprioceptive, interoceptive, and arousal hindbrain networks share a developmental requirement for Math1. *Proc. Natl. Acad. Sci. U. S. A.* **106**, 22462–22467 (2009).

37. Ruffault, P.-L. *et al.* The retrotrapezoid nucleus neurons expressing Atoh1 and Phox2b are essential for the respiratory response to CO2. *eLife* **4**, (2015).

38. Dubreuil, V. *et al.* A human mutation in Phox2b causes lack of CO2 chemosensitivity, fatal central apnea, and specific loss of parafacial neurons. *Proc. Natl. Acad. Sci. U. S. A.* **105**, 1067–1072 (2008).

39. Thoby-Brisson, M. *et al.* Genetic identification of an embryonic parafacial oscillator coupling to the preBötzinger complex. *Nat. Neurosci.* **12**, 1028–1035 (2009).

40. Ramanantsoa, N. *et al.* Breathing without CO2 Chemosensitivity in Conditional Phox2b Mutants. *J. Neurosci.* **31**, 12880–12888 (2011).

41. Jacquin, T. D. *et al.* Reorganization of pontine rhythmogenic neuronal networks in Krox-20 knockout mice. *Neuron* **17**, 747–758 (1996).

42. Di Lascio, S. *et al.* Structural and functional differences in PHOX2B frameshift mutations underlie isolated or syndromic congenital central hypoventilation syndrome. *Hum. Mutat.* **39**, 219–236 (2018).

43. Cleary, C. M., James, S., Maher, B. J. & Mulkey, D. K. Disordered breathing in a Pitt-Hopkins syndrome model involves Phox2b-expressing parafacial neurons and aberrant Nav1.8 expression. *Nat. Commun.* **12**, 5962 (2021).

44. Dosumu-Johnson, R. T., Cocoran, A. E., Chang, Y., Nattie, E. & Dymecki, S. M. Acute perturbation of Pet1-neuron activity in neonatal mice impairs cardiorespiratory homeostatic recovery. *eLife* **7**, (2018).

45. Sauber-Schatz, E. K., Sappenfield, W. M. & Shapiro-Mendoza, C. K. Comprehensive review of sleep-related sudden unexpected infant deaths and their investigations: Florida 2008. *Matern. Child Health J.* **19**, 381–90 (2015).

46. Martin, R. J. & Wilson, C. G. Apnea of prematurity. *Compr. Physiol.* **2**, 2923–2931 (2012).

47. Palkovic, B., Marchenko, V., Zuperku, E. J., Stuth, E. A. E. & Stucke, A. G. Multi-Level Regulation of Opioid-Induced Respiratory Depression. *Physiology* **35**, 391–404 (2020).

48. Ramirez, J.-M. *et al.* Neuronal mechanisms underlying opioid-induced respiratory depression: our current understanding. *J. Neurophysiol.* **125**, 1899–1919 (2021).

49. Oliveira, L. M., Baertsch, N. A., Moreira, T. S., Ramirez, J.-M. & Takakura, A. C. Unraveling the Mechanisms Underlying Irregularities in Inspiratory Rhythm Generation in a Mouse Model of Parkinson's Disease. *J. Neurosci.* **41**, 4732–4747 (2021).

50. Benarroch, E. E., Schmeichel, A. M., Low, P. A. & Parisi, J. E. Depletion of ventromedullary NK-1 receptor-immunoreactive neurons in multiple system atrophy. *Brain J. Neurol.* **126**, 2183–2190 (2003).

51. Tsuboi, Y. *et al.* Neurodegeneration involving putative respiratory neurons in Perry syndrome. *Acta Neuropathol. (Berl.)* **115**, 263–268 (2008).

52. Benarroch, E. E. Brainstem in multiple system atrophy: clinicopathological correlations. *Cell. Mol. Neurobiol.* **23**, 519–526 (2003).

53. Takeda, S. *et al.* Opioid action on respiratory neuron activity of the isolated respiratory network in newborn rats. *Anesthesiology* **95**, 740–749 (2001).

54. David, C. K. *et al.* Single cell transcriptome sequencing of inspiratory neurons of the preBötzinger complex in neonatal mice. *Sci. Data* **9**, 457 (2022).

55. Hayes, J. A. *et al.* Transcriptome of neonatal preBötzinger complex neurones in Dbx1 reporter mice. *Sci. Rep.* **7**, 8669 (2017).

56. Kallurkar, P. S. *et al.* Transcriptomes of electrophysiologically recorded Dbx1-derived respiratory neurons of the preBötzinger complex in neonatal mice. *Sci. Rep.* **12**, 2923 (2022).

57. Macosko, E. Z. *et al.* Highly Parallel Genome-wide Expression Profiling of Individual Cells Using Nanoliter Droplets. *Cell* **161**, 1202–1214 (2015).

58. Chung, N. C. & Storey, J. D. Statistical significance of variables driving systematic variation in high-dimensional data. *Bioinforma. Oxf. Engl.* **31**, 545–554 (2015).

59. van der Maaten, L. & Hinton, G. Visualizing data using t-SNE. *J. Mach. Learn. Res.* **9**, 2579–2605 (2008).

60. Becht, E. *et al.* Dimensionality reduction for visualizing single-cell data using UMAP. *Nat. Biotechnol.* **37**, 38–44 (2019).

61. Ester, M., Kriegel, H.-P., Sander, J. & Xu, X. A density-based algorithm for discovering clusters in large spatial databases with noise. in *KDD-96 Proceedings, AAAI* (AAAI, 1996).

62. McDavid, A. *et al.* Data exploration, quality control and testing in single-cell qPCR-based gene expression experiments. *Bioinforma. Oxf. Engl.* **29**, 461–467 (2013).

63. Picardo, M. C. D. *et al.* Trpm4 ion channels in pre-Bötzinger complex interneurons are essential for breathing motor pattern but not rhythm. *PLoS Biol.* **17**, e2006094 (2019).

64. Bankhead, P. *et al.* QuPath: Open source software for digital pathology image analysis. *Sci. Rep.* **7**, 16878 (2017).

65. Gray, P. A., Rekling, J. C., Bocchiaro, C. M. & Feldman, J. L. Modulation of respiratory frequency by peptidergic input to rhythmogenic neurons in the preBötzinger complex. *Science* **286**, 1566–1568 (1999).

66. Stornetta, R. L. *et al.* A group of glutamatergic interneurons expressing high levels of both neurokinin-1 receptors and somatostatin identifies the region of the pre-Bötzinger complex. *J. Comp. Neurol.* **455**, 499–512 (2003).

67. Guyenet, P. G. & Wang, H. Pre-Bötzinger neurons with preinspiratory discharges ‘in vivo’ express NK1 receptors in the rat. *J. Neurophysiol.* **86**, 438–446 (2001).

68. Guyenet, P. G., Sevigny, C. P., Weston, M. C. & Stornetta, R. L. Neurokinin-1 receptor-expressing cells of the ventral respiratory group are functionally heterogeneous and predominantly glutamatergic. *J. Neurosci. Off. J. Soc. Neurosci.* **22**, 3806–3816 (2002).

69. Tan, W. *et al.* Reelin demarcates a subset of pre-Bötzinger complex neurons in adult rat. *J. Comp. Neurol.* **520**, 606–619 (2012).

70. Takakura, A. C., Barna, B. F., Cruz, J. C., Colombari, E. & Moreira, T. S. Phox2b-expressing retrotrapezoid neurons and the integration of central and peripheral chemosensory control of breathing in conscious rats: Phox2b-expressing retrotrapezoid neurons and the chemoreflex. *Exp. Physiol.* **99**, 571–585 (2014).

71. Takakura, A. C. *et al.* Selective lesion of retrotrapezoid Phox2b-expressing neurons raises the apnoeic threshold in rats: NK1 receptors in RTN. *J. Physiol.* **586**, 2975–2991 (2008).

72. Mellen, N. M., Janczewski, W. A., Bocchiaro, C. M. & Feldman, J. L. Opioid-induced quantal slowing reveals dual networks for respiratory rhythm generation. *Neuron* **37**, 821–826 (2003).

73. Bachmutsky, I., Wei, X. P., Kish, E. & Yackle, K. Opioids depress breathing through two small brainstem sites. *eLife* **9**, e52694 (2020).

74. Li, P. *et al.* The peptidergic control circuit for sighing. *Nature* **530**, 293–297 (2016).

75. Shi, Y. *et al.* Neuromedin B Expression Defines the Mouse Retrotrapezoid Nucleus. *J. Neurosci. Off. J. Soc. Neurosci.* **37**, 11744–11757 (2017).

76. Abbott, S. B. G., Stornetta, R. L., Coates, M. B. & Guyenet, P. G. Phox2b-expressing neurons of the parafacial region regulate breathing rate, inspiration, and expiration in conscious rats. *J. Neurosci. Off. J. Soc. Neurosci.* **31**, 16410–16422 (2011).

77. Abdala, A. P. L., Rybak, I. A., Smith, J. C. & Paton, J. F. R. Abdominal expiratory activity in the rat brainstem-spinal cord in situ: patterns, origins and implications for respiratory rhythm generation. *J. Physiol.* **587**, 3539–3559 (2009).

78. Vann, N. C., Pham, F. D., Hayes, J. A., Kottick, A. & Del Negro, C. A. Transient suppression of Dbx1 preBötzinger interneurons disrupts breathing in adult mice. *PloS One* **11**, e0162418 (2016).

79. Vann, N. C., Pham, F. D., Dorst, K. E. & Del Negro, C. A. Dbx1 Pre-Bötzinger Complex Interneurons Comprise the Core Inspiratory Oscillator for Breathing in Unanesthetized Adult Mice. *eNeuro* **5**, (2018).

80. Banzett, R. B., Lansing, R. W. & Binks, A. P. Air Hunger: A Primal Sensation and a Primary Element of Dyspnea. *Compr. Physiol.* **11**, 1449–1483 (2021).

81. Parshall, M. B. *et al.* An Official American Thoracic Society Statement: Update on the Mechanisms, Assessment, and Management of Dyspnea. *Am. J. Respir. Crit. Care Med.* **185**, 435–452 (2012).